

# Data Management in Practice

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# Policy landscape

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UNESCO's Recommendation on Open Science

EU

- Open Science Policy
- Directives
- European Research Council
- Horizon Europe
- EOSC

Swedish research bills 2016 & 2020

- Transition to open research data implemented by 2026
- Government assignments to KB & VR

SUHF national roadmap for open science and University policies

Lund Declaration on Maximising the Benefits of Research Data

*National guidelines for open science (KB)*

*Open Science*  
**FAIR**

*“FAIR [...] open data sharing  
should become the default  
[...]”*

*“As open as possible,  
as closed as necessary”*



# VR - Swedish Research Council

## Swedish Research Council recommends open access to research data

research process. Already existing data that have only been used in their original form and that are already managed and made accessible by another actor are not covered by this recommendation.

### Metadata should also be published with open access

Both research data and data describing research data (known as metadata) should be published with open access. If there are obstacles to publishing research data, the focus should in the first instance be on making metadata openly accessible on the internet. In this way, users can find information on what research data exists, even when there are obstacles to open publication, for example lack of a suitable publication platform or technical limitations that prevent all data from being published.

### Publication according to the FAIR principles

Publication of research data can be done using various digital platforms, for example via the higher education institution where the research is conducted or via other relevant national and/or international portals, infrastructures and similar organisations and platforms. The publication of research data shall always be based on the FAIR principles.

### The Swedish Research Council's recommendation on data management according to FAIR

The Swedish Research Council recommends that the research data produced through research are managed according to the FAIR principles, clarified via the criteria developed by the Swedish Research Council to achieve FAIR data.

The FAIR principles should be implemented taking into account applicable legislation, and, as far as is possible and applicable, based on the technical, organisational and/or discipline-specific preconditions that apply.

The recommendations relates in the first instance to research data (and metadata) financed by public funds that can be published with open access, but the application of the FAIR principles can be made broader than this, and be used also for research data that cannot be published entirely openly. The recommendation on data management according to FAIR is overarching, and aims to create a common starting point for the implementation of FAIR data management.

***[...] The publication of research data shall always be based on the FAIR principles.[...]***

***The Swedish Research Council's recommendation on data management according to FAIR***

***The Swedish Research Council recommends that the research data produced through research are managed according to the FAIR principles, clarified via the criteria developed by the Swedish Research Council to achieve FAIR data. [...]***

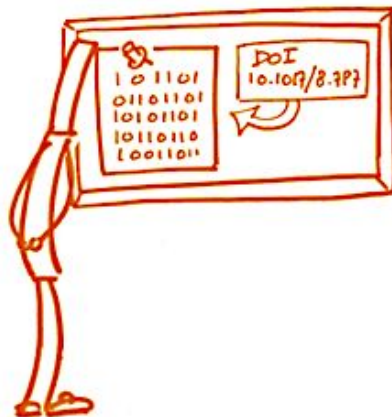


# Sharing - Stay FAIR

## FAIR DATA PRINCIPLES



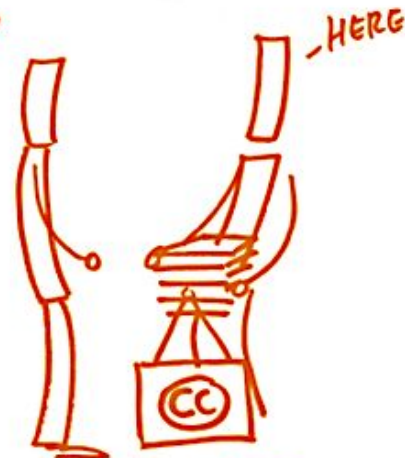
FINDABLE



ACCESSIBLE



INTEROPERABLE



REUSABLE

# Keeping data in a good shape

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## **Organise your files in a structured way**

- Use a file naming-convention
- Separate raw data from processed data
- Use simple README text files to describe content of folders

## **Use standard, non-proprietary, file formats**

## **Have a plan for storing your data**

## **Have a strategy for backing up your data**

## **Stick to available standards for metadata (i.e. data about the data)**

- Learn from repositories where the type of data can be deposited
- Make sure to collect relevant metadata as soon as the information is available
- Store the metadata where it is easy to find

## **Document changes to your project**

- Consider using a version control system

Year 0



The ideas!

Year 1



Data collection!

Year 2



Results!

Year 3



Paper out!

Finished!

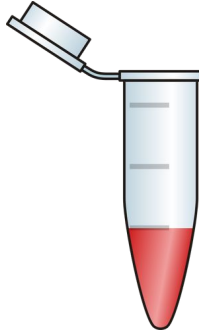
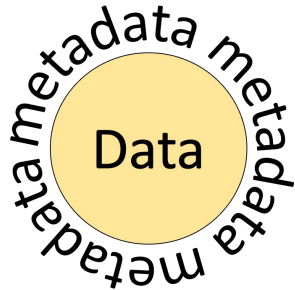
Oh, I need to publish  
the data also? Ok.

I can do that!

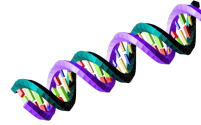
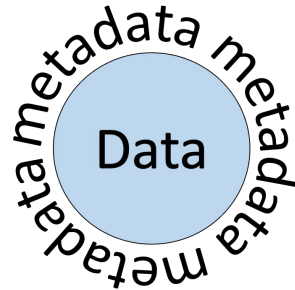
# What is metadata?



Source: [Openclipart](#)

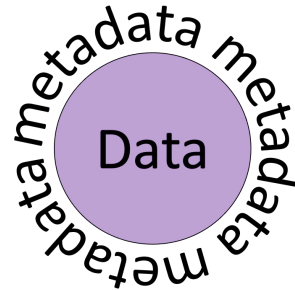


Source: [Openclipart](#)



**.fastq**

Source: [Openclipart](#)



Source: [Publicdomainpictures](#)



What samples did you use,  
and how are they described?  
(Sample metadata)

What machines did you  
run your analysis on?  
(Technical metadata)

Which files to publish, and  
where are they?



Year 0



Year 1



Year 2



Year 3



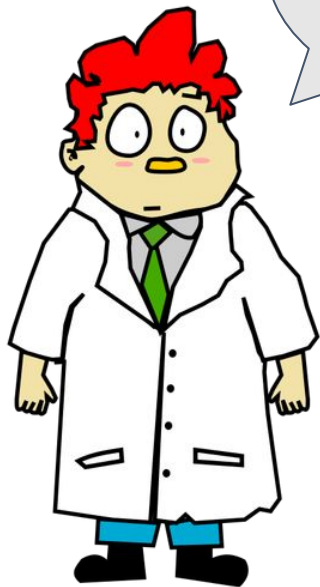
Sample metadata?  
What is it? Where is  
it saved? Did I even  
save it at all?  
(3 years ago)

Technical  
metadata? What is  
that, and who do I  
ask?  
(2 years ago)

The data files?  
They are ALL here!  
At least I think so...  
(1-2 years ago)

# The Nice Reality

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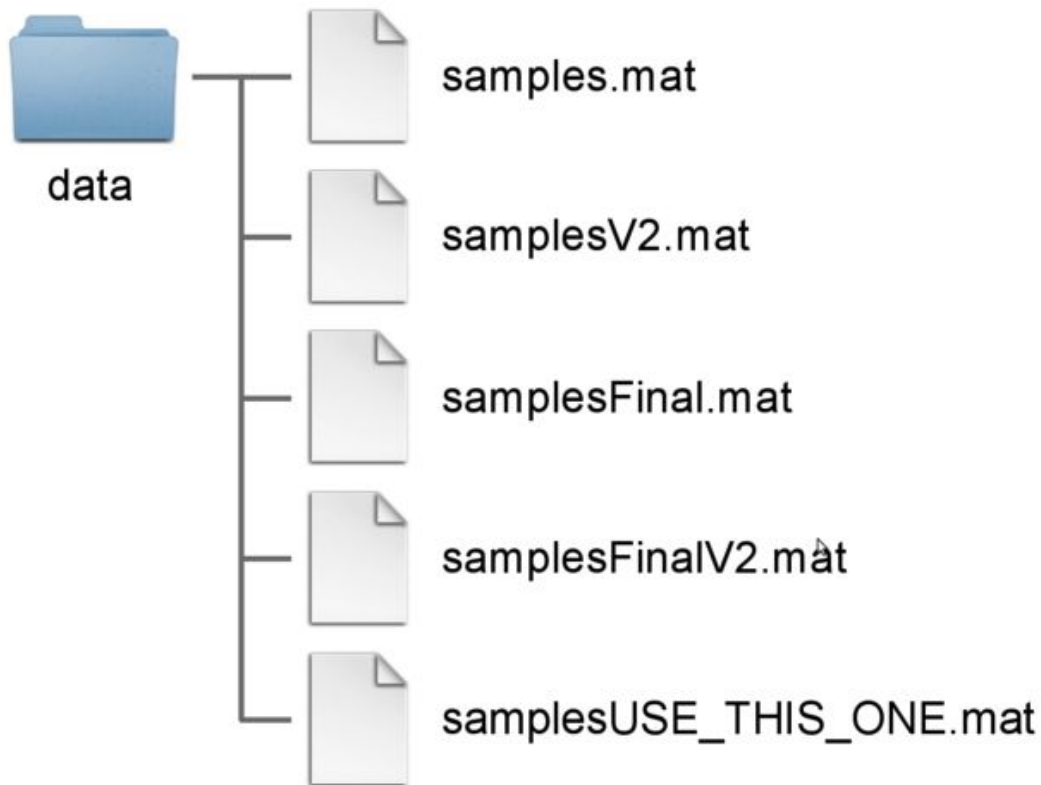


**What do I have?**

- Spreadsheet with sample information
- Electronic notebook with lab protocol
- Delivery report from sequencing facility
- A bioinformatic analysis report
- A bunch of data files somewhere

**How do I describe so that others can understand?**

Source: [Openclipart](#)



# Why submit to a repository?

*“The data is available upon request”*

Many reasons:

- Open Science & FAIR
- Reproducibility
- Trail of evidence
- 3rd party access
- Archival purposes
- Publication of paper requires it



Digitalbevaring.dk

Credit: Illustration from Digitalbevaring.dk / Jørgen Stamp  
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# What research outputs should be submitted?

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- **Raw data:** straight from the instrument eg fastq, bam, cram
- **Processed data:** normalization, removal of outliers, expression measurements, statistics
- **Metadata:** minimum information to reproduce the data, sample information, precise protocols
- **Code:** software code that is needed to re-run analyses

# Types of repositories

## Domain-specific:

- Best choice - long-term plan, typically free, maximum reach
- E.g. [European Nucleotide Archive](#), [European Genome Phenome Archive](#), [ArrayExpress](#), [PRIDE](#)

## General purpose:

- Second best – long-term plan, might cost (now or in future), good reach but less specific in metadata → more difficult for future users to judge if a dataset will be useful
- E.g. [Zenodo](#), [SciLifeLab Data Repository \(Figshare\)](#), [Dryad](#)

## In-house/institutional

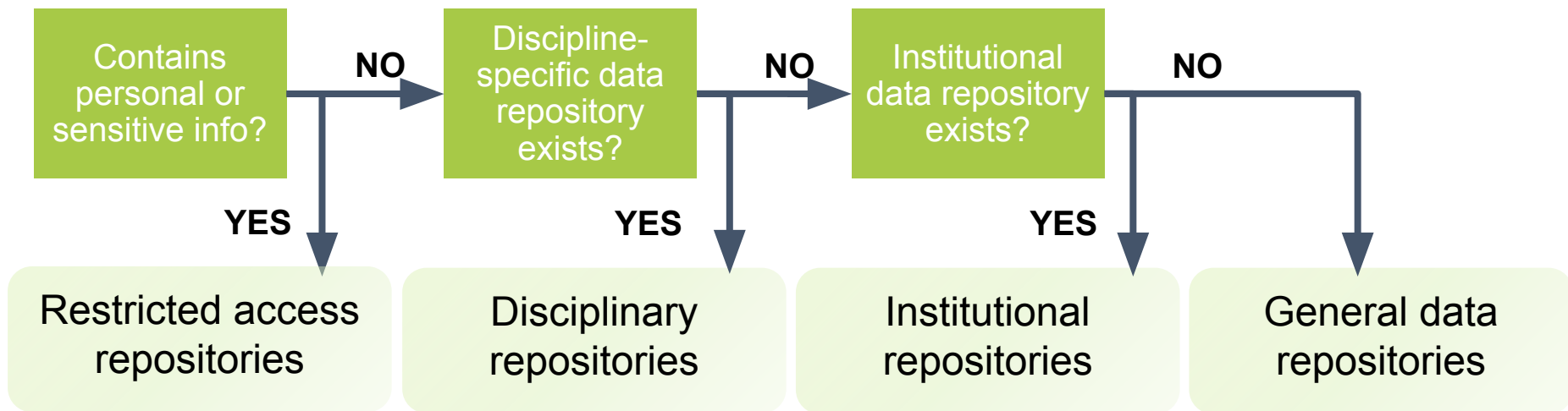
- For archive/backup purpose mainly, might cost, limited reach unless also published in a data catalogue

Domain-specific

General  
purpose

In-house

# Selecting a Data Repository



# What about sensitive data?

Data regarded as special category data under GDPR may be possible to share under **controlled access**.

Controlled access means that researchers only will be granted access after a formal application procedure.

- The [European Genome-phenome Archive \(EGA\)](#) is a repository for archiving and sharing sensitive personal data from biomedical research projects
- [FEGA Sweden](#) – the Swedish node of the Federated EGA which is working to become operational.

If you cannot deposit the data in a repository: create at least a record describing the data (a "metadata-record") in e.g. [SciLifeLab Data Repository](#).

